
Quantitative & Machine-Learning Models

The complete mathematical specification

This volume is the exhaustive mathematical reference for every quantitative and machine-learning model in AI Analyst. It specifies the cross-sectional gradient-boosted alpha model, the factor-structured risk models, the dual portfolio optimizers, the walk-forward out-of-sample backtest with Fama–French benchmarking, the rolling robust factor regression, the macro-cycle regime models (PCA/DFM, Gaussian HMM, temporal multimodal VAE, and regime-conditioned factor IC), and the deterministic valuation and scoring models — each with its objective, estimator, algorithm, and the exact warehouse tables and code paths it runs on.

Companion volumes: System Architecture & Operations, qlib Integration, and SEC & EDINET Metrics Mapping & Reconciliation.

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Part I Foundations

1 Scope and notation

AI Analyst answers two different questions with two different families of model. The *valuation* models (Part IV) answer “*what is this business worth?*” deterministically, from projected cash flows and a cost of capital. The *quantitative* models (Parts II–III) answer “*what should I expect from this stock, and how should I size it?*” statistically, from a learned cross-sectional return forecast, a forward covariance, and a portfolio optimizer, conditioned on an estimated macro regime. This document is the mathematical specification of both families.

Notation. $\mathbf{w} \in \mathbb{R}^N$ portfolio weights; $\boldsymbol{\mu} \in \mathbb{R}^N$ expected (annualized) returns; $\Sigma \in \mathbb{R}^{N \times N}$ return covariance; r_f the risk-free rate; $\mathbf{1}$ the ones vector; $\odot$ the elementwise (Hadamard) product; z a cross-sectional z -score. Time subscripts t index months unless stated otherwise. Annualization uses 252 trading days for daily series and 12 for monthly series. A “hat” ($\hat{\cdot}$) denotes an estimate or a model prediction.

Every model is designed to *degrade gracefully*: if a trained artifact, an optional dependency (`hmmlearn`, `PyTorch`, a MIP solver), or a warehouse table is absent, the code returns a defined fallback rather than failing, so the application always runs. Those fallbacks are stated with each model.

2 The data substrate: a point-in-time fundamentals panel

Every cross-sectional model consumes the same object: a monthly panel of standardized fundamentals aligned *point-in-time* (PIT) so that no signal ever uses data that was not yet public.

2.1 Point-in-time alignment

A fundamental value with fiscal-period end p is stamped usable only at

$$\tau(p) = \text{monthend}(p + 90 \text{ days}), \quad (1)$$

a conservative proxy for the filing lag. In SQL the alignment is `(date_trunc('month', period_end + INTERVAL '90 days') + INTERVAL '1 month - 1 day')`. The alpha model (§3) and the regime-IC engine (§9.6) share this one loader, so the research signal and the production forecast see an identical, lookahead-safe history.

2.2 Feature families and cross-sectional standardization

Features are importance-ranked standardized fundamentals (`fact_metrics_{us,jp}`, keeping importance ≤ 2) drawn from four families:

Family	Representative factors (substring patterns)
value	book-to-market, earnings yield, FCF yield, dividend, EV/·, valuation multiples
quality	return-on-·, ROE/ROA, margins, accruals, cash conversion, asset turnover
growth	revenue/earnings growth, investment, capex, R&D, sales change
market_factor	market beta, volatility, momentum, residual/abnormal return

For each factor j , name i , and month t , the raw value $x_{i,t}^{(j)}$ is *cross-sectionally* standardized and winsorized (mirroring qlib’s `CSZScoreNorm`):

$$z_{i,t}^{(j)} = \text{clip}\left(\frac{x_{i,t}^{(j)} - \bar{\mu}_t^{(j)}}{\hat{\sigma}_t^{(j)}}, -3, 3\right), \quad \bar{\mu}_t^{(j)} = \frac{1}{N_t} \sum_i x_{i,t}^{(j)}, \quad \hat{\sigma}_t^{(j)} = \text{std}_i x_{i,t}^{(j)}. \quad (2)$$

Standardizing within each date makes factors comparable across names and dates and robust to per-metric scale; a zero-variance cross-section maps to 0 (the neutral value). A missing feature is imputed to the cross-sectional mean, i.e. $z = 0$.

2.3 Label

The supervised label is the forward one-month (or three-month) total return,

$$y_{i,t} = r_{i,t \rightarrow t+h}, \quad h \in \{1, 3\} \text{ months.} \quad (3)$$

The panel is built as a qlib MultiIndex frame — rows (datetime, instrument), columns (feature | label) — consumed directly by `DataHandlerLP.from_df`. JP fundamentals carry a .T suffix that is stripped so fundamentals join prices. Thin cross-sections (< 30 names) are dropped as unstable for ranking and normalization.

Design

The panel is the single contract between the warehouse and the models: a monthly, cross-sectional, lookahead-safe frame of z -scored fundamentals plus a forward return. Expected returns from the alpha model are therefore *monthly* forecasts — annualize by $\times 12$ before feeding a portfolio optimizer.

Part II The qlib quant desk

3 Cross-sectional alpha model (return prediction)

Code: `api/quant/qlib_alpha.py`, `api/quant/qlib_data.py`, `api/quant/alpha_signal.py`; built on qlib's `LGBModel` (LightGBM).

3.1 Objective

Learn a function $f: \mathbf{z} \mapsto \hat{y}$ mapping the standardized cross-sectional feature vector of a name to its expected forward return. The prediction $\hat{y} = f(\mathbf{z})$ is the *alpha signal*: a per-name, per-month score interpreted as an expected forward return.

3.2 Gradient-boosted decision trees

The model is an additive ensemble of M regression trees with learning rate η ,

$$F_M(\mathbf{z}) = \sum_{m=1}^M \eta h_m(\mathbf{z}), \quad h_m \text{ a regression tree.} \quad (4)$$

Under the squared-error objective $\mathcal{L} = \sum_i \frac{1}{2} (y_i - F(\mathbf{z}_i))^2$, boosting stage m fits the new tree h_m to the negative functional gradient — here simply the residual — with unit Hessian:

$$g_i = -\left. \frac{\partial \mathcal{L}}{\partial F} \right|_{F_{m-1}} = y_i - F_{m-1}(\mathbf{z}_i), \quad h_i = \frac{\partial^2 \mathcal{L}}{\partial F^2} = 1. \quad (5)$$

LightGBM grows each tree *leaf-wise* over a histogram approximation of the features. Splitting a node with instance set I into I_L, I_R yields the gain

$$\text{Gain} = \frac{1}{2} \left[\frac{(\sum_{i \in I_L} g_i)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{(\sum_{i \in I_R} g_i)^2}{\sum_{i \in I_R} h_i + \lambda} - \frac{(\sum_{i \in I} g_i)^2}{\sum_{i \in I} h_i + \lambda} \right] - \gamma, \quad (6)$$

where λ is the L_2 leaf regularizer and γ the per-split penalty. Trees capture non-linearities and factor *interactions* that a linear factor model cannot represent.

Shipped hyperparameters (DEFAULT_LGB_PARAMS)

<code>loss = mse</code>	<code>learning_rate = 0.03</code>	<code>num_leaves = 31</code>	<code>max_depth = 6</code>
<code>colsample_bytree = 0.8</code>	<code>subsample = 0.8</code>	<code>subsample_freq = 1</code>	<code>min_child_samples = 50</code>
<code>lambda_l1 = 1.0</code>	<code>lambda_l2 = 1.0</code>	<code>num_boost_round = 500</code>	<code>early_stopping_rounds = 40</code>

Regularization (L_1/L_2 leaf penalties, shallow depth, feature/row subsampling) plus early stopping on a held-out validation segment control overfitting in a setting with a low signal-to-noise ratio.

3.3 Training, splitting, and persistence

The panel is split *chronologically* by unique dates into contiguous train / validation / test segments (defaults 70/15/15). Training runs the real `qlib DatasetH` pipeline; the `AlphaLGB` subclass overrides `fit` to attach LightGBM's early-stopping and evaluation callbacks directly, so training runs with plain LightGBM and needs no `qlib.init` or metric-tracking backend. Prediction runs the stored booster on any reindexed feature matrix — including a single live month — so scoring never rebuilds a `DatasetH`. Artifacts (the booster plus metadata: feature identity/order, label horizon, train range, IC) are serialized per jurisdiction and hot-reloaded on file `mtime`.

For the live cross-section the model builds a *features-only* panel (`require_label=False`) over a short lookback and scores the newest month: the current month has no realized forward return, so a label join would drop exactly the cross-section we want to score.

The split is contiguous by default, which leaks whenever the label horizon exceeds one month: an h -month label observed at the last training month is not realized until h months later, i.e. inside the validation block. `time_segments` therefore accepts an `embargo_months` gap that purges that many months from the end of each earlier segment; the research loop of §7 sets it to the label horizon by default. Figures computed without it are optimistic and are not comparable with the purged walk-forward of §7.3.

3.4 Evaluation: the information coefficient

Predictive power is measured per month by the cross-sectional correlation between prediction and realized return:

$$\text{IC}_t = \text{Corr}(\hat{y}_{\cdot,t}, y_{\cdot,t}) \text{ (Pearson)}, \quad \text{RankIC}_t = \rho_{\text{Spearman}}(\hat{y}_{\cdot,t}, y_{\cdot,t}), \quad (7)$$

summarized by the mean IC and the information ratio of the signal, $\text{ICIR} = \overline{\text{IC}} / \text{std}(\text{IC})$ (and its rank analogue). A monthly equity signal with $\overline{\text{RankIC}} \approx 0.05\text{--}0.10$ is usable. The *in-sample* test segment of the shipped US/JP models reaches $\overline{\text{RankIC}} \approx 0.08\text{--}0.14$; the *out-of-sample* walk-forward figure (§6) is materially lower and is the number used for any performance claim.

Note that ICIR as defined above is a bare mean/standard-deviation ratio, not the annualized information ratio the name usually denotes; the comparable monthly-panel figure carries a $\sqrt{12}$ scaling (§7.4). Both are reported by `evaluate()`.

4 Factor-structured risk models (forward covariance)

Code: `api/quant/qlib_risk.py` (statistical) and `api/quant/risk.py` (Ledoit–Wolf, Fama–French).

A portfolio optimizer needs a forward covariance Σ that is well conditioned and invertible. A raw $N \times N$ sample covariance from a short daily window is noisy and near-singular. All three risk models replace it with a structured, lower-parameter estimator. Each is selectable at runtime.

4.1 Statistical factor covariance (qlib StructuredCovEstimator)

Returns are modeled as driven by K latent statistical factors,

$$X = B F^T + U, \quad \text{Cov}(X) = F \text{Cov}(B) F^T + \text{diag}(\text{Var}(U)), \quad (8)$$

where $X \in \mathbb{R}^{T \times N}$ is the return panel, $F \in \mathbb{R}^{N \times K}$ the factor *exposures* (principal components; factor analysis is also supported), $B \in \mathbb{R}^{T \times K}$ the factor *returns* (component scores), and U the idiosyncratic residual. Writing $\Sigma_B = \text{Cov}(B)$ and $\mathbf{d} = \text{Var}(U)$,

$$\Sigma = F \Sigma_B F^\top + \text{diag}(\mathbf{d}) \quad (9)$$

— a low-rank-plus-diagonal shrinkage of the sample covariance. The number of factors is clamped to $K = \min(K_0, N - 1, T - 2)$ for safety. Estimation is on *raw* returns (the estimator’s percentage-rescaling is disabled, keeping variances in return units); Σ_B and $\mathbf{d}$ are annualized by $\times 252$, and Σ is rebuilt from the annualized decomposition, symmetrized, and given a 10^{-10} ridge for numerical positive-definiteness. Forward volatility per name is $\sigma_i = \sqrt{\Sigma_{ii}}$. The decomposition $(F, \Sigma_B, \mathbf{d})$ is consumed directly by the enhanced-indexing optimizer (§5.3).

4.2 Ledoit–Wolf shrinkage (constant-correlation target)

An analytic shrinkage of the MLE sample covariance S toward a constant-correlation target F . With variances s_{ii} , standard deviations $\sigma_i = \sqrt{s_{ii}}$, and mean pairwise correlation $\bar{r} = \frac{2}{N(N-1)} \sum_{i < j} s_{ij} / (\sigma_i \sigma_j)$, the target is $F_{ii} = s_{ii}$, $F_{ij} = \bar{r} \sigma_i \sigma_j$, and

$$\Sigma^* = \delta F + (1 - \delta) S, \quad \delta = \text{clip}\left(\frac{\pi/\gamma}{T}, 0, 1\right), \quad (10)$$

where $\gamma = \|S - F\|_F^2$ measures target misspecification and $\pi = \sum_{i,j} [\frac{1}{T} \sum_t (x_{ti} x_{tj} - s_{ij})^2]$ is the summed asymptotic variance of the entries of S . The optimal intensity δ trades sample noise against target bias; the result is annualized $\times 252$.

4.3 Fama–French fundamental covariance

A named-factor analogue of the statistical model:

$$\Sigma = B F_{\text{ann}} B^\top + D, \quad (11)$$

with B the per-name FF loadings (`fact_factor_loadings`, §8), $F_{\text{ann}} = 252 \cdot \text{Cov}(\text{factor returns})$ estimated over the lookback, and $D = \text{diag}(\text{residual_vol}_i^2)$ from the regression meta. Model ids `ff3`, `ff5`, `ff5_plus_mom` select the factor set (FF6 = FF5 + momentum). Names with missing loadings fall back to their sample diagonal with a surfaced warning; an unavailable factor model degrades to Ledoit–Wolf.

5 Portfolio optimization (dual backend)

Code: `api/quant/qlib_optimize.py` dispatches at runtime to a *native* optimizer (`api/quant/optimize.py`) or any qlib optimizer; both consume the same μ, Σ and return a uniform `PortfolioSolution`.

5.1 Native optimizer: multi-term mean–variance utility

The native backend maximizes a mean–variance utility with factor-tilt, turnover, and concentration penalties:

$$\max_{\mathbf{w}} \quad \mathbf{w}^\top \boldsymbol{\mu} - \frac{1}{2} \lambda \mathbf{w}^\top \Sigma \mathbf{w} - \gamma_f \|S \odot a \odot (B^\top \mathbf{w} - \mathbf{t})\|_2^2 - \gamma_{\text{to}} \|\mathbf{w} - \mathbf{w}_0\|_1 - \gamma_c \|\mathbf{w}\|_2^2, \quad (12)$$

subject to full investment $\mathbf{1}^\top \mathbf{w} = 1$, per-name bounds $l_i \leq w_i \leq u_i$, a gross cap $\sum_i |w_i| \leq G$, a short-leg cap $\sum_i \max(-w_i, 0) \leq \rho \sum_i |w_i|$, sector caps $\mathbf{a}_s^\top \mathbf{w} \leq c_s$, an annualized volatility cap $\mathbf{w}^\top \Sigma \mathbf{w} \leq \sigma_{\text{max}}^2$, and factor caps/ranges $|\mathbf{b}_j^\top \mathbf{w}| \leq \text{cap}_j$. Here λ is risk aversion; γ_f penalizes deviation of the portfolio’s factor exposures $B^\top \mathbf{w}$ from targets $\mathbf{t}$ (scaled by S , gated by the active mask a); γ_{to} penalizes turnover against the current book $\mathbf{w}_0$; and γ_c penalizes concentration.

The problem is solved by sequential least-squares programming (SLSQP), which handles the nonlinear volatility and factor constraints natively and converges quickly at portfolio scale ($N \lesssim 100$). SLSQP is given the analytic gradient

$$\nabla = -\boldsymbol{\mu} + \lambda \Sigma \mathbf{w} + 2\gamma_f B(S^2 \odot a \odot (B^T \mathbf{w} - \mathbf{t})) + \gamma_{to} \text{sign}(\mathbf{w} - \mathbf{w}_0) + 2\gamma_c \mathbf{w}. \quad (13)$$

Cardinality (mixed-integer program). When a maximum position count $\|\mathbf{w}\|_0 \leq K$ is requested, the problem becomes a mixed-integer program with binary inclusion indicators $y_i \in \{0, 1\}$:

$$\sum_i y_i \leq K, \quad \begin{cases} 0 \leq w_i \leq u_i y_i, & \text{long-only,} \\ -u_i y_i \leq w_i \leq u_i y_i, & \text{long/short,} \end{cases} \quad w_i \geq f y_i \text{ (floor on included names),} \quad (14)$$

retaining the convex volatility, sector, and factor constraints. It is solved via `cvxpy` over the first available MIP backend (`HiGHS` → `SCIP` → `CBC` → `GLPK_MI` → `MOSEK`); if none is installed the route surfaces a clear “MIP unavailable” error rather than silently dropping the constraint.

Diagnostics and frontier. The solver reports factor exposures $B^T \mathbf{w}$, marginal risk contributions $(\Sigma \mathbf{w}) \odot \mathbf{w} / \sqrt{\mathbf{w}^T \Sigma \mathbf{w}}$, the effective number of names $1 / \sum_i w_i^2$, and the covariance condition number; it also sweeps λ over a fixed grid to trace a mini efficient frontier.

5.2 qlib optimizers (long-only, fully invested)

All satisfy $\mathbf{1}^T \mathbf{w} = 1$, $\mathbf{w} \geq 0$:

Backend	Objective
GMV (min variance)	$\min_{\mathbf{w}} \mathbf{w}^T \Sigma \mathbf{w}$
MVO (mean–variance)	$\max_{\mathbf{w}} \lambda \tilde{r}^T \mathbf{w} - \mathbf{w}^T \Sigma \mathbf{w} \quad (\tilde{r} \text{ scaled to } \Sigma\text{'s volatility})$
RP (risk parity)	choose $\mathbf{w}$ s.t. $w_i (\Sigma \mathbf{w})_i = w_j (\Sigma \mathbf{w})_j \quad \forall i, j$
Inverse-vol	$w_i = \frac{1/\sigma_i}{\sum_j 1/\sigma_j}, \quad \sigma_i = \sqrt{\Sigma_{ii}}$

Risk parity equalizes the risk contributions $RC_i = w_i (\Sigma \mathbf{w})_i$, which sum to total variance $\sum_i RC_i = \mathbf{w}^T \Sigma \mathbf{w}$; GMV/MVO/RP are solved with `scipy.optimize` and admit optional turnover (δ) and L_2 (α) terms.

5.3 Enhanced indexing (benchmark-relative)

Given benchmark weights $\mathbf{w}_b$, active weights $\mathbf{d} = \mathbf{w} - \mathbf{w}_b$, and active factor exposure $\mathbf{v} = \mathbf{d}^T F$, the enhanced-indexing optimizer maximizes active return minus active (tracking-error) risk under bounded name and factor deviations:

$$\max_{\mathbf{w}} \mathbf{d}^T \mathbf{r} - \lambda (\mathbf{v}^T \Sigma_B \mathbf{v} + \mathbf{d}^T \text{diag}(\mathbf{d}_u) \mathbf{d}) \quad \text{s.t.} \quad \mathbf{w} \geq 0, \mathbf{1}^T \mathbf{w} = 1, \|\mathbf{w} - \mathbf{w}_0\|_1 \leq \delta, |\mathbf{d}| \leq b_{\text{dev}}, |\mathbf{v}| \leq f_{\text{dev}}, \quad (15)$$

where $\mathbf{d}_u = \text{Var}(U)$ is the specific variance. It is a convex quadratic program consuming the statistical-covariance decomposition $(F, \Sigma_B, \mathbf{d}_u)$ of §4 directly, solved by `cvxpy/ECOS`; on solver failure it degrades to the benchmark weights with a warning.

5.4 Reported portfolio statistics

For a solution $\mathbf{w}^*$ the desk reports $\mathbb{E}[r]_{\text{ann}} = \mathbf{w}^{*T} \boldsymbol{\mu}$, $\sigma_{\text{ann}} = \sqrt{\mathbf{w}^{*T} \Sigma \mathbf{w}^*}$, Sharpe = $(\mathbb{E}[r]_{\text{ann}} - r_f) / \sigma_{\text{ann}}$, and the factor exposures and risk contributions above.

6 Walk-forward out-of-sample backtest

Code: `api/quant/qlib_backtest.py`. A cross-sectional signal backtest that is out-of-sample by construction, benchmarked against Fama–French.

6.1 Walk-forward predictions

Let $t_0 < t_1 < \dots < t_T$ be the panel months, restricted to an investable universe (market cap $\geq \$2\text{B}$) with each month's forward returns winsorized to the 1st/99th cross-sectional percentile (removing micro-cap data artifacts). For each test month t_i with $i \geq m_{\min}$ (default $m_{\min} = 12$; refit every $q = 3$ months) a booster g_{t_i} is trained on *all prior* months $\{t_j : t_j < t_i\}$ and predicts $\hat{\alpha}_{k,t_i} = g_{t_i}(\mathbf{z}_{k,t_i})$. No month is ever scored by a model that trained on it. The expensive walk-forward predictions are cached separately from portfolio construction, so re-tuning top- k or long/short re-ranks instantly.

6.2 Portfolio construction

Ranking names by $\hat{\alpha}$ each month, the equal-weight books realize

$$r_t^L = \frac{1}{k} \sum_{i \in \text{top-}k} r_{i,t}, \quad r_t^{\text{LS}} = \frac{1}{k} \sum_{i \in \text{top-}k} r_{i,t} - \frac{1}{k} \sum_{i \in \text{bot-}k} r_{i,t}, \quad (16)$$

for the long-only and long/short strategies respectively; a month is skipped if it has fewer than $\max(k, 10)$ (long) or $2k$ (long/short) investable names.

6.3 Performance and risk metrics

With monthly returns $\{r_t\}_{t=1}^n$, equity $E_t = \prod_{s \leq t} (1 + r_s)$, and monthly risk-free $r_{f,t}$:

$$R_{\text{ann}} = \left(\prod_t (1 + r_t) \right)^{12/n} - 1, \quad \sigma_{\text{ann}} = \text{std}(r_t) \sqrt{12}, \quad (17)$$

$$\text{Sharpe} = \frac{R_{\text{ann}} - R_{\text{ann}}^f}{\sigma_{\text{ann}}}, \quad \text{Sortino} = \frac{R_{\text{ann}} - R_{\text{ann}}^f}{\text{std}(r_t : r_t < 0) \sqrt{12}}, \quad (18)$$

$$\text{MaxDD} = \min_t \left(\frac{E_t}{\max_{s \leq t} E_s} - 1 \right), \quad \text{Hit} = \frac{1}{n} \sum_t \mathbf{1}[r_t > 0]. \quad (19)$$

6.4 Fama–French benchmarking and factor attribution

The benchmark is the FF *market* total return $r_t^m = (\text{Mkt-RF})_t + (\text{RF})_t$, compounded from daily `fact_fama_french` to monthly and aligned to $\{r_t\}$. Relative to it,

$$\text{excess} = R_{\text{ann}} - R_{\text{ann}}^m, \quad \text{TE} = \text{std}(r_t - r_t^m) \sqrt{12}, \quad \text{IR} = \frac{\text{excess}}{\text{TE}}, \quad \beta = \frac{\text{Cov}(r, r^m)}{\text{Var}(r^m)}. \quad (20)$$

To separate genuine alpha from factor exposure, the strategy's excess return is regressed on the FF five factors plus momentum,

$$r_t - r_{f,t} = \alpha + \beta_{\text{Mkt}} f_t^{\text{Mkt}} + \beta_{\text{SMB}} f_t^{\text{SMB}} + \beta_{\text{HML}} f_t^{\text{HML}} + \beta_{\text{RMW}} f_t^{\text{RMW}} + \beta_{\text{CMA}} f_t^{\text{CMA}} + \beta_{\text{Mom}} f_t^{\text{Mom}} + \varepsilon_t. \quad (21)$$

OLS on the design $A = [\mathbf{1} \ X]$ gives $\hat{\theta} = (A^\top A)^{-1} A^\top y$, residual variance $\hat{s}^2 = \|\varepsilon\|^2 / (n - K - 1)$, coefficient covariance $\hat{s}^2 (A^\top A)^{-1}$, the alpha t -statistic $t_\alpha = \hat{\alpha} / \text{se}(\hat{\alpha})$, annualized alpha $\alpha_{\text{ann}} = (1 + \hat{\alpha})^{12} - 1$, and $R^2 = 1 - \|\varepsilon\|^2 / \sum_t (y_t - \bar{y})^2$.

Interpretation

The walk-forward design is what makes the backtest honest: because every month is scored by a model that never saw it, the out-of-sample $\bar{\text{RankIC}}$ and t_α are the figures used for any claim about the signal. They are markedly weaker than the in-sample test-segment numbers of §3 — the expected outcome of removing look-ahead, and the reason performance is always quoted from this path.

Part III Factor and macro-regime models

7 Agentic model research

Code: `api/quant/research/`. The estimators in this section replace the single fixed specification of §3 with a searched one. The search is driven by four LLM agents (their orchestration is documented in the *qlib Integration* volume); what follows is the *statistics* they steer.

7.1 The searched specification

A `TrainingSpec` is a tuple θ collecting every choice that was previously hard-coded: the sample filters, the outlier treatment, the standardization estimator, the imputation rule, the feature block, the split geometry, the model family and its hyperparameters. Writing $\mathcal{D}$ for the raw panel, a round evaluates

$$\theta \mapsto (\hat{g}_\theta, \mathcal{M}(\theta)), \quad \hat{g}_\theta = \text{fit}(\Pi_\theta(\mathcal{D})), \quad (22)$$

where Π_θ is the preprocessing map below and $\mathcal{M}$ the metric battery of §7.4. Agents propose a *patch* to θ ; a deterministic validator whitelists every key and clamps every value before it can reach a fit, so the reachable set is exactly the specified product space.

7.2 Robust cross-sectional estimators

The incumbent standardization (§2.2) uses the cross-sectional mean and standard deviation, then clips at ± 3 . Both moments are computed *from the sample that contains the outlier*, so a single extreme value shifts the centre and inflates the scale for every other name in that month; the clip then protects the model from the outlier but not the cross-section from its influence. Three alternatives are searchable.

Winsorization by order statistics. For a month with N_t observations and a tail fraction q , let $k = \max(1, \lfloor qN_t \rfloor)$ and let $x_{(1)} \leq \dots \leq x_{(N_t)}$ be the order statistics. Each of the k most extreme observations per tail is replaced by the next one in:

$$\tilde{x}_{i,t} = \text{clip}(x_{i,t}, x_{(k+1)}, x_{(N_t-k)}). \quad (23)$$

This is deliberately *not* $\text{clip}(x, \hat{Q}_q, \hat{Q}_{1-q})$ with interpolated sample quantiles. An interpolated quantile is estimated from the contaminated sample, so on a thin cross-section the outlier drags its own bound up to meet it — with $N_t = 60$ and $q = 0.01$ a planted value of 500 pulled $\hat{Q}_{0.99}$ to ≈ 206 and the clip did essentially nothing. The order-statistic form has no such feedback, and the floor $k \geq 1$ keeps it meaningful when $qN_t < 1$.

Robust standardization. With med_t the cross-sectional median and $\text{MAD}_t = \text{med}_i |x_{i,t} - \text{med}_t|$,

$$z_{i,t} = \text{clip}\left(\frac{x_{i,t} - \text{med}_t}{1.4826 \text{MAD}_t}, -c, c\right), \quad (24)$$

the constant 1.4826 making the MAD a consistent estimator of σ under normality. Both the location and the scale have a 50% breakdown point, against 0% for the mean/standard-deviation pair.

Rank transform. Discarding magnitude entirely and keeping only the ordering the model is scored on:

$$z_{i,t} = 2 \left(\frac{\text{rank}_i(x_{i,t})}{N_t} - \frac{1}{2} \right) \in [-1, 1]. \quad (25)$$

Imputation and neutralization. A missing feature may be filled at the cross-sectional *median* rather than the mean (the incumbent's $z = 0$), or the row dropped. Features may additionally be residualized within each month against sector dummies and log size,

$$\tilde{\mathbf{z}}_t = \mathbf{z}_t - Z_t(Z_t^\top Z_t)^{-1} Z_t^\top \mathbf{z}_t, \quad (26)$$

one least-squares solve per month for the whole feature block, which asks whether a factor says anything beyond which industry a company is in.

7.3 Purged walk-forward evaluation

The incumbent split is contiguous with no gap, so with an h -month label the last h training months share realized future returns with the validation block. Let E be the embargo, defaulting to $E = h$. Training at prediction month t_i is restricted to

$$\mathcal{T}_{t_i} = \{t_j : t_j \leq t_i - E\}, \quad (27)$$

so a model predicting month t has seen only labels already realized by $t - E$ — the constraint a live deployment actually operates under. The same gap purges the train/validation seam of the inner split used for early stopping.

The embargo is not cosmetic

On the shipped US `forward_12m` model the purged walk-forward rank-IC of the incumbent specification is *negative*. Figures quoted from the unpurged single-block split are not comparable with it and are optimistic by construction.

7.4 The metric battery

Organized by quality attribute rather than by convenience, after Lewis et al.’s observation that ML evaluation overwhelmingly reports predictive accuracy and little else, and that testing across attributes surfaces a wider class of defects.

Out-of-sample R^2 . Both conventions are reported, because they diverge and quoting one without saying which is how implausible figures arise:

$$R_{\text{oos},0}^2 = 1 - \frac{\sum (y - \hat{y})^2}{\sum y^2}, \quad R_{\text{oos},\mu}^2 = 1 - \frac{\sum (y - \hat{y})^2}{\sum (y - \bar{y})^2}. \quad (28)$$

The zero-benchmarked form is the return-prediction standard: the historical mean is itself a fitted quantity, and benchmarking against it flatters the model.

Scaled information ratio. $\text{ICIR} = \overline{\text{IC}} / \text{std}(\text{IC})$ is a bare ratio, not the annualized information ratio the name suggests. On a monthly panel the comparable figure is

$$\text{ICIR}^{\text{ann}} = \frac{\overline{\text{IC}}}{\text{std}(\text{IC})} \sqrt{12}, \quad (29)$$

roughly $3.46\times$ larger. Both are reported; the unscaled key retains its meaning for existing consumers.

Uncertainty. A percentile bootstrap over the per-month RankIC_t series gives a 95% interval for $\overline{\text{RankIC}}$. An interval spanning zero is treated as *no demonstrated skill* regardless of the point estimate, and blocks promotion.

Ranking quality. Beyond the top-minus-bottom decile spread and its t -statistic, the *monotonicity* $\rho_{\text{Spearman}}(d, \bar{r}_d)$ between decile index d and its mean realized return distinguishes a genuine cross-sectional ordering from a tail effect in a handful of names — a large spread with low monotonicity is the latter.

Implementability and hygiene. Turnover is the fraction of the top- k book replaced each month. The FF5+momentum regression of the long/short series separates alpha from factor exposure; a small t_α beside a large R^2 means the strategy is a factor tilt. The *factor-neutral* rank-IC re-scores the signal after residualizing it cross-sectionally on the names’ Fama–French betas.

Stability. The train-minus-OOS IC gap is the overfitting tell. Feature-attribution rankings are themselves checked for stability across refits (top- k Jaccard and rank correlation) before any conclusion is drawn from them, and the dispersion of the selected model complexity across windows is reported: hyperparameters that swing between windows indicate a low signal-to-noise ratio rather than a tuning opportunity.

7.5 Perturbation robustness rating

Adapted from the causal rating method of Lakkaraju et al. for time-series forecasters. The fitted model is held *fixed* and the data it is served is degraded, so the procedure needs no access to the training pipeline and is auditable by a third party. Five perturbations map to concrete warehouse failure modes: feature values dropped (missing or late filings), halved (unit-scale and concept-mapping errors), shifted one month stale (point-in-time slippage), the newest month withheld (reporting delay), and fat-tail contamination of the label (corporate actions).

For perturbation P the relative degradation is

$$\delta_P = \frac{\overline{\text{RankIC}}_\emptyset - \overline{\text{RankIC}}_P}{|\overline{\text{RankIC}}_\emptyset|}. \quad (30)$$

GICS industry is the confounder: a perturbation may appear damaging only because it concentrates in an over-represented industry. With a single low-cardinality categorical confounder, exact stratification is not an approximation of propensity matching but the exact version of what it approximates, so the effect is directly standardized,

$$\bar{e}^{\text{std}} = \frac{1}{|G|} \sum_{g \in G} \bar{e}_g, \quad (31)$$

and the gap $||\bar{e}| - |\bar{e}^{\text{std}}||/|\bar{e}|$ is reported as the share of the apparent effect attributable to industry composition.

Aggregation is *worst case* rather than average — a desk choosing a model needs the floor — and maps to an ordinal rating

$$\text{rating} = \begin{cases} 1 \text{ (robust)} & \max_P \delta_P < 0.25, \\ 2 \text{ (moderate)} & 0.25 \leq \max_P \delta_P < 0.60, \\ 3 \text{ (fragile)} & \text{otherwise.} \end{cases} \quad (32)$$

Why a rating and not another number

The rating separates models that a single skill statistic ranks identically — or backwards. On a controlled panel an over-parameterized fit posts a *higher* in-sample-flavoured rank-IC (0.78 vs 0.34) than a regularized one while rating 3 against 1: it memorizes, and small realistic data defects destroy it.

7.6 Sub-population skill

The headline is an average over a heterogeneous universe and can hide a sign flip. Skill is therefore reported per bucket under two cuts that ask different questions: GICS sector and industry group (*where* does this work, in the terms a PM allocates by) and within-date quintiles of the point-in-time Fama–French loadings $\beta_{\text{mkt}}, \beta_{\text{smb}}, \beta_{\text{hml}}, \beta_{\text{mom}}$ (*what kind of company* does it work on, in the terms a risk model prices). Each bucket reports $\overline{\text{RankIC}}$, its t -statistic, R_{oos}^2 and the decile spread; buckets below a minimum name count are computed but flagged, because a headline resting on one is a finding rather than a result.

On the live US panel the two cuts are informative in exactly this way: an incumbent specification whose overall rank-IC is -0.013 carries $+0.040$ in Industrials, and its skill concentrates in the lowest market-beta quintile.

7.7 Champion selection and combination

Candidates surviving validation compete against an equal-weight combination of themselves, formed by averaging *within-date* percentile ranks,

$$\hat{\alpha}_{i,t}^{\text{ens}} = \frac{1}{M} \sum_{m=1}^M \frac{\text{rank}_i(\hat{\alpha}_{i,t}^{(m)})}{N_t}, \quad (33)$$

which is scale-free and so combines a GBDT and an elastic net on equal terms. The persisted ensemble instead sums member predictions standardized by their own training moments, because the serving path scores a bare array with no date grouping available; the two agree on ordering closely enough that selection remains meaningful.

8 Rolling Fama–French factor regression

Code: `xbrl_sec/sec/sources/factor_model.py`. Per ticker, over a rolling 252-day window (stepped every 21 trading days, minimum 126 observations), excess returns are regressed on the FF factors of the chosen model (FF3/FF4/FF5/FF6):

$$r_{i,t} - r_{f,t} = \alpha_i + \sum_k \beta_{i,k} f_{k,t} + \varepsilon_{i,t}. \quad (34)$$

Factor series are region-resolved (US, JP, Developed, Emerging) from the available Ken French datasets and converted to decimals; the dependent window is winsorized to the 1st/99th percentile.

8.1 Robust estimation (Huber IRLS)

Rather than plain OLS, coefficients are estimated by iteratively reweighted least squares with Huber weights, which down-weight outlier days. Over a small number of iterations, with scale $\hat{s} = \text{median}(|e|)/0.6745$ and $u = |e|/\hat{s}$,

$$\hat{\beta}^{(k+1)} = (X^T W^{(k)} X)^{-1} X^T W^{(k)} y, \quad w_i = \begin{cases} 1, & u_i \leq \delta, \\ \delta/u_i, & u_i > \delta, \end{cases} \quad \delta = 1.345. \quad (35)$$

8.2 HAC (Newey–West) standard errors

Standard errors are heteroskedasticity- and autocorrelation-consistent with Bartlett weights over $L = 5$ lags:

$$\widehat{\text{Var}}(\hat{\beta}) = (X^T X)^{-1} \hat{S} (X^T X)^{-1}, \quad \hat{S} = \hat{\Gamma}_0 + \sum_{\ell=1}^L w_{\ell} (\hat{\Gamma}_{\ell} + \hat{\Gamma}_{\ell}^T), \quad w_{\ell} = 1 - \frac{\ell}{L+1}, \quad (36)$$

with $\hat{\Gamma}_{\ell} = \sum_t (x_t \varepsilon_t)(x_{t-\ell} \varepsilon_{t-\ell})^T$.

8.3 Stored diagnostics and derived series

Each window stores α (annualized $\times 252$), the betas, t -statistics, p -values, and standard errors, plus R^2 , adjusted $R^2 = 1 - (1 - R^2) \frac{n-1}{n-k}$, F -statistic and its p -value, RMSE, residual volatility $\hat{\sigma}_{\varepsilon} \sqrt{252}$ (the idiosyncratic D of §4.3), the Durbin–Watson statistic $\sum_t (\varepsilon_t - \varepsilon_{t-1})^2 / \sum_t \varepsilon_t^2$, and the design condition number. A composite `quality_score` $\in \{0, \dots, 5\}$ counts how many of five gates a window passes (adj. $R^2 > 0.20$; F $p < 0.05$; $n \geq 200$; condition number < 30 ; $1.5 \leq \text{DW} \leq 2.5$). Between refits, per-day model-implied returns $\hat{r}_{i,t} = \alpha_i/252 + \sum_k \beta_{i,k} f_{k,t} + r_{f,t}$ are written to `fact_factor_implied_returns`. These loadings feed the WACC (§10.2) and the FF structured covariance (§4.3).

9 Macro-cycle regime models

Code: `xbrl_sec/sec/cycle/`. The regime stack has three layers: a multimodal feature set, a dimensionality-reduction *baseline* that produces latent cycle factors, and two *regime models* (HMM and VAE) that run on top, each mapped to interpretable business-cycle phases and then linked to forecasting through regime-conditioned factor IC.

9.1 Feature construction

`cycle/features.py` assembles a monthly, point-in-time, multimodal feature matrix across three modalities — *macro* (growth, inflation, rates, liquidity, credit), *market*, and *fundamental* — each observation stamped with its true availability date. Features are standardized with an *expanding* (walk-forward) mean and standard deviation, $z_{f,t} = (x_{f,t} - \mu_{f,\leq t}) / \sigma_{f,\leq t}$ (minimum 12 observations), so no future information leaks into any month’s normalization.

9.2 Baseline: PCA / dynamic-factor compression

`cycle/baselines.py` compresses the (mean-centered) feature matrix to K latent cycle factors via the SVD $X = U\Sigma V^T$, taking the leading components factors $= U_{:,1:K} \Sigma_{1:K}$ with explained-variance ratios $\sigma_k^2 / \sum \sigma^2$. The `dfm` (dynamic-factor) variant additionally smooths the factors with an exponentially weighted moving average (span 3), approximating the persistence of a dynamic factor model. The factors are written to `fact_cycle_state_monthly.latent_cycle` and are the input to the two regime models.

9.3 Phase calibration and labeling

Latent factors have arbitrary sign and order, so `cycle/phase.py` anchors them to interpretable phases. It (i) selects the factor most correlated with an external stress/recession proxy from the macro store and orients it so “high = stress”; (ii) maps the factor to a rolling stress *percentile* s_t over a 120-month window; and (iii) thresholds s_t into four phases {contraction, late_cycle, mid_expansion, early_expansion}. For US models the thresholds are calibrated against the NBER recession chronology (via USREC’s trough convention) by maximizing balanced accuracy $\frac{1}{2}(\text{TPR} + \text{TNR})$ over a grid; labels are then smoothed (EWMA) and passed through a minimum-phase-duration filter that dissolves runs shorter than three months. Phase confidence is a bounded function of distance from the 0.5 percentile, $c_t = \text{clip}(1.5 |s_t - 0.5| + 0.45, 0.45, 0.90)$.

9.4 Gaussian Hidden Markov Model

`cycle/hmm.py` fits a Gaussian HMM (default $K = 4$ states, full covariance) over the latent cycle factors $\mathbf{x}_t$:

$$\mathbf{x}_t \mid s_t=k \sim \mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k), \quad A_{ij} = \mathbb{P}(s_t=j \mid s_{t-1}=i), \quad \pi_k = \mathbb{P}(s_1=k). \quad (37)$$

The parameters $\{\pi, A, \boldsymbol{\mu}_k, \Sigma_k\}$ are fit by Baum–Welch (the EM algorithm for HMMs). The forward–backward recursions define $a_t(k) = \mathbb{P}(\mathbf{x}_{1:t}, s_t=k)$ and $b_t(k) = \mathbb{P}(\mathbf{x}_{t+1:T} \mid s_t=k)$, from which the smoothed state posterior is

$$\gamma_t(k) = \mathbb{P}(s_t=k \mid \mathbf{x}_{1:T}) = \frac{a_t(k) b_t(k)}{\sum_{k'} a_t(k') b_t(k')}. \quad (38)$$

The per-month confidence is $\max_k \gamma_t(k)$ and the MAP state path is recovered by Viterbi. States are mapped to phases by ordering their mean stress score (§9.3). If `hmmlearn` is unavailable, a deterministic k -means-style assignment with empirical transition counts produces the same output shape.

9.5 Temporal multimodal variational autoencoder

`cycle/vae.py` is an optional VAE with encoder $q_\phi(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\phi(\mathbf{x}), \text{diag } \sigma_\phi^2(\mathbf{x}))$, decoder $p_\theta(\mathbf{x} \mid \mathbf{z})$, and prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, I)$. It is trained by maximizing the evidence lower bound with a temporal

smoothness penalty:

$$\mathcal{L}(\theta, \phi) = \underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x} | \mathbf{z})]}_{\text{reconstruction}} - \beta \underbrace{\text{KL}(q_\phi(\mathbf{z} | \mathbf{x}) \| p(\mathbf{z}))}_{\text{regularization}} - \eta \sum_t \|\boldsymbol{\mu}_{\phi,t} - \boldsymbol{\mu}_{\phi,t-1}\|^2, \quad (39)$$

using the reparameterization $\mathbf{z} = \boldsymbol{\mu}_\phi + \boldsymbol{\sigma}_\phi \odot \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I)$ (in the shipped configuration $\beta = 0.05$, $\eta = 0.02$). The latent dimensions are split into interpretable *growth* / *inflation* / *rates-liquidity* / *credit-stress* / *market* / *fundamentals* factors. The smoothed reconstruction-error stress percentile is thresholded into phases exactly as in §9.3, and per-modality reconstruction MSE attributes stress to macro vs. market vs. fundamentals. If PyTorch is absent, a PCA surrogate produces the same persisted output shape.

9.6 Regime-conditioned factor IC

`cycle/ic.py` is the link from the regime model to forecasting. For each fundamental factor m , *conditioned on the regime*, it computes the rank information coefficient against forward returns over each month and horizon:

$$\text{IC}_{\text{regime}}(m) = \rho_{\text{Spearman}}(x_{m,i}, r_i^{\text{fwd}} | \text{regime}). \quad (40)$$

Two variants are stored to `fact_equity_factor_ic_regime`: a *hard-assigned* IC (grouping by the MAP phase label) and a *probability-weighted* IC that weights each observation by the regime posterior $\gamma_t(k)$ (soft assignment), using the weighted Spearman correlation of the value and forward-return ranks. The result answers *which factors have actually predicted returns in conditions like today's*, and drives the scanner's factor-family weights (§11.1).

From regime to score: the one live consumption path

`ic_weights.family_weights` reads the *latest date's* regime-IC rows, aggregates per-metric $|\text{IC}|$ to family-level emphasis $e_{\text{fam}} = |\text{IC}|_{\text{fam}}$ (normalized to sum to one over value / growth / quality), and caches it. The scanner then re-splits its fundamental score budget in proportion to e_{fam} : the interest score leans harder on whichever factor family the current regime says is most predictive, and reverts *exactly* to the fixed weights when the IC table is unavailable. The learned alpha model itself is *regime-agnostic* — it learns from factor values, and the regime enters forecasting through the score weights, not (currently) as a model feature.

Part IV Deterministic valuation and scoring

10 Valuation models

Code: `ai_analyst/dcf_engine.py`, `committee/wacc.py`, `committee/valuation.py`, `committee/segments.py`. Deterministic, on a 7-year explicit horizon (DCF_HORIZON_YEARS). The models are intentionally transparent so the equity-research “why” of every number reads off the projected statements; the LLM committee supplies only the bounded assumptions, and the math below turns them into figures.

10.1 Discounted cash flow (unlevered FCFF)

For each forecast year $i = 1, \dots, 7$:

$$\text{Rev}_i = \text{Rev}_{i-1}(1 + g_i), \quad \text{EBIT}_i = \text{Rev}_i \cdot m, \quad (41)$$

$$\text{NOPAT}_i = \text{EBIT}_i - \max(\text{EBIT}_i, 0) \tau, \quad \text{FCF}_i = \text{NOPAT}_i + \text{D\&A}_i - \text{Capex}_i - \Delta\text{NWC}_i, \quad (42)$$

with D&A, Capex, and ΔNWC taken as fractions of revenue (D&A% from the latest historical ratio). Terminal value uses Gordon growth, and the enterprise/equity bridge is

$$\text{TV} = \frac{\text{FCF}_7(1 + g_\infty)}{\text{WACC} - g_\infty}, \quad \text{EV} = \sum_{i=1}^7 \frac{\text{FCF}_i}{(1 + \text{WACC})^i} + \frac{\text{TV}}{(1 + \text{WACC})^7}, \quad (43)$$

Equity = EV – NetDebt, per share = Equity/shares. A 5×5 sensitivity grid varies WACC by ± 200 bp and g_∞ by ± 100 bp (a growth $\times$ WACC grid is also produced around the base case).

10.2 Cost of capital (WACC)

Cost of equity is built from the FF factor betas (§8) and long-run premia $\Pi_k = 252 \cdot \overline{f_k}$:

$$R_e^{\text{CAPM}} = r_f + \beta_{\text{Mkt}} \text{ERP}, \quad R_e^{\text{FF5}} = r_f + \sum_k \beta_k \Pi_k, \quad (44)$$

with r_f the 10Y Treasury (FRED:DGS10); both are clamped to $[r_f + 2\%, r_f + 12\%]$. Cost of debt is $R_d = r_f + \text{spread}$, where the spread comes from a Damodaran-style synthetic-rating table keyed on interest coverage EBIT/Interest, blended with the realized interest/debt rate when that looks plausible. Then

$$\text{WACC} = \frac{E}{V} R_e + \frac{D}{V} R_d (1 - \tau), \quad V = E + D, \quad (45)$$

E market cap, D total financial debt, clamped to $[6\%, 16\%]$. The headline uses R_e^{CAPM} (the market convention and a stable discount rate); the multi-factor R_e^{FF5} is reported alongside as an exhibit. For the sum-of-the-parts DCF, `segment_wacc` shifts the equity beta and adds a growth-risk premium so higher-growth segments carry a higher discount rate.

10.3 Reverse DCF, scenarios, sum-of-the-parts, triangulation

Reverse DCF solves, by bisection, the flat growth g^* (or steady margin m^*) such that $\text{PerShare}(g^*) = \text{price}$ — i.e. what the market is currently pricing in. *Scenario weighting* runs the DCF once per upside/base/downside scenario (sharing one deterministic share count and net debt) and combines the per-share values V_s with probabilities $\sum_s p_s = 1$ into $\text{FV} = \sum_s p_s V_s$. *Sum-of-the-parts* values each reportable segment on its own growth/margin and a segment-specific WACC, with segment $\text{FCF} \approx \text{NOPAT} \times \text{conversion}$, and sums the segment enterprise values. *Triangulation* builds a football field from the SOTP range (declared primary), the consolidated DCF scenario range, and peer-multiples-implied values, reporting implied upside $\text{FV}/\text{price} - 1$.

11 Scoring and ranking models

11.1 Value + sentiment interest score

Code: `api/routers/screener_agent.py`. Each component is min–max normalized across the shortlist to $[0, 1]$ (v_{PE} inverted — cheaper is better), plus a normalized alpha v_α from the qlib model:

$$\text{score} = w_\alpha v_\alpha + (1 - w_\alpha) \sum_k w_k v_k, \quad \text{interest} = 100 \cdot \text{score}, \quad (46)$$

with $w_\alpha \approx 0.40$ (dominant when the alpha model covers the shortlist). The fundamental sub-weights w_k (FCF-yield, P/E, growth, MD&A tone, news) start from availability-branched defaults and are re-split between the *value* and *growth* blocks by the current regime’s factor IC (§9.6): with family emphases $e_{\text{val}}, e_{\text{grw}}$, the fundamental budget is allocated $\propto e_{\text{val}} : e_{\text{grw}}$. With no model or IC available it reduces exactly to the fixed-weight composite.

11.2 Group relative-value composite

Code: `ai_analyst/committee/group.py`. For each metric k (P/E, EV/EBITDA, P/B, FCF yield, dividend, revenue growth/CAGR, gross/operating margin), a cross-sectional z -score $z_{k,i} = (x_{k,i} - \bar{x}_k)/\hat{\sigma}_k$ and a signed weight w_k (negative for cheaper-is-better multiples). Loss-making multiples (negative P/E, EV/EBITDA, P/B) are excluded from their metric’s population rather than ranked as “cheap”. The composite is

$$\text{composite}_i = \sum_k w_k z_{k,i}, \quad (47)$$

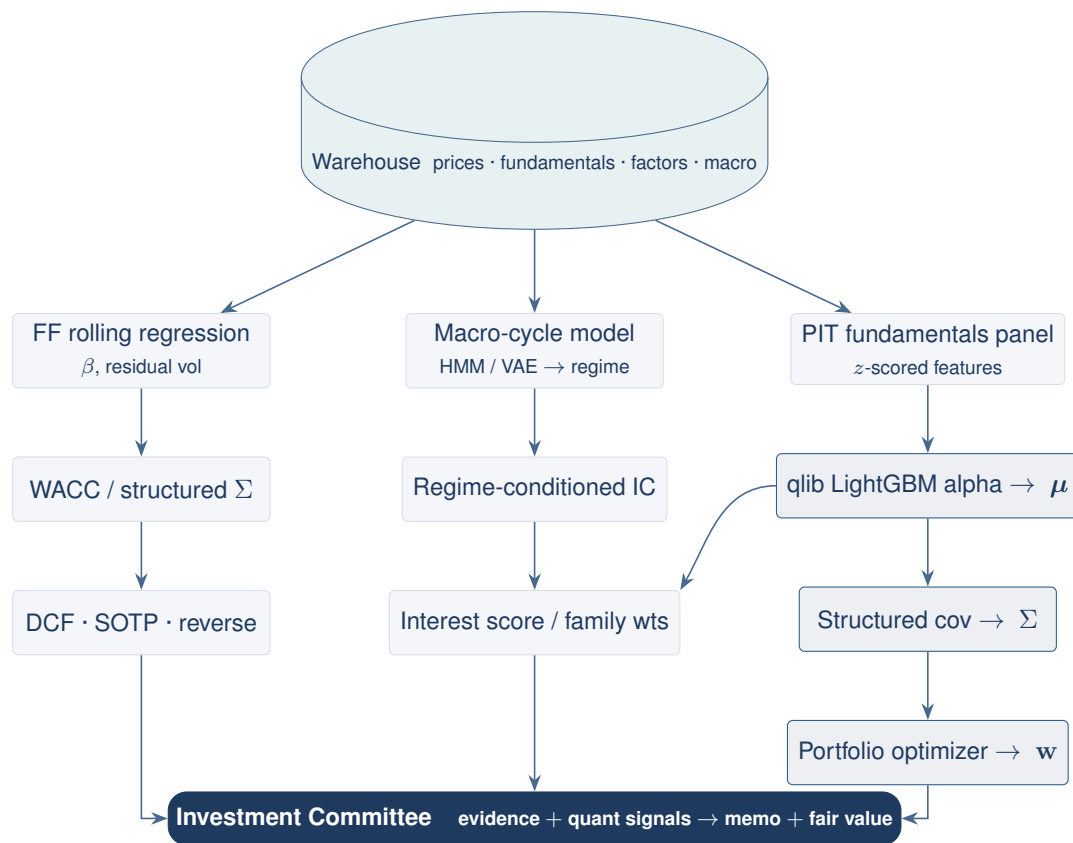


Figure 1. The model map. Statistical, macro, and valuation paths converge on the committee, which produces a reasoned valuation with its evidence rather than a single label.

ordered descending; terciles map to attractive / fair / expensive. The per-name audit trail ($x \rightarrow z \rightarrow \times w \rightarrow \text{contribution}$) sums exactly to the composite. An optional LLM narrative may be attached; the ranking itself is deterministic.

11.3 MD&A management tone

An LLM reads an excerpt of each name's latest MD&A and returns a structured tone in $[-1, 1]$, a positive/neutral/negative guidance label, and risk flags; the tone feeds v_{tone} in §11.1.

Part V How the models connect

12 The model map

The three quantitative paths and the valuation path meet at the investment committee. The valuation path answers what a business is worth; the quant path answers what to expect and how to size it; the macro-cycle path conditions the scoring on the regime; and the committee fuses the deterministic evidence and the quant signals into a reasoned, auditable verdict.

Independent, systematic research for educational purposes. Estimates are model outputs, not investment advice.